

Positioning in DECT-2020 NR with Proactive Measurement Selection

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Abstract—This paper addresses efficient positioning and user device tracking in Internet-of-Things (IoT) networks with particular focus on the new DECT-2020 New Radio standard – the first non-cellular 5G technology standard in the world. Stemming from fundamental performance requirements of IoT networks, for example, related to energy consumption, latency, and reliability, it is important to utilize available radio resources efficiently, while avoiding redundant transmissions and signaling. In this paper, we propose a tracking-based positioning solution with a proactive measurement selection, in which positioning measurements are selected based on Fisher information. The proposed method, accompanied with computationally attractive processing solution, is able to sort and select potential positioning measurements based on achievable minimum positioning error. The provided numerical results, considering both range and angle measurements, show that the proposed method outperforms the traditional signal-to-noise ratio based measurement selection approach, and enables positioning with a smaller number of measurements.

Index Terms—positioning, DECT-2020 NR, internet-of-things, non-cellular 5G, Fisher information, extended Kalman filter

I. INTRODUCTION

To accommodate growing demand on Internet-of-Things (IoT) based services, the new Digital Enhanced Cordless Telecommunications - 2020 New Radio (DECT-2020 NR) [1], [2] was introduced as part of the official International Telecommunication Union (ITU) fifth generation (5G) standards in 2022 [3]. The DECT-2020 NR is the first non-cellular 5G technology, designed especially for IoT use cases with support for 5G performance requirements regarding ultra reliable low latency communications (URLLC) and massive machine type communications (mMTC).

Positioning plays an important role in various IoT and wireless sensor network (WSN) systems, as well as related applications and location-based services (LBS) [4]. DECT-2020 NR-based positioning is not yet widely studied due to its technological novelty, but research works focusing on positioning with other IoT systems, or IoT-like scenarios, can be found, for example, in [4]–[9]. In [4], different IoT-based positioning approaches are compared, and classified in a IoT infrastructure applicable framework, while also considering aspects of power consumption and energy efficiency. Furthermore, possible positioning methods for IoT systems are comprehensively discussed in [5], [9], including also comparison between positioning capabilities of different IoT technologies.

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In this paper, we study positioning and user device tracking in IoT and WSN like systems using angle-of-arrival (AoA) and time-of-arrival (ToA)-based range measurements obtained from known anchor devices, while specifically focusing on the capabilities and physical-layer of the new DECT-2020 NR standard. Furthermore, we provide and derive fundamental performance bounds, and propose a novel Fisher information based measurement selection method to choose the most important anchor measurements for minimizing the positioning error. The proposed measurement selection approach has not been earlier presented in the literature, but prior research regarding range and AoA based measurements have been earlier considered, e.g., in [7], [8]. Moreover, a signal-to-noise ratio (SNR)-based approach for measurement and reference anchor selection with time-difference-of-arrival (TDoA) measurements is studied and described in [10], serving as a reference scheme.

Through the proposed proactive approach, it is possible to reduce the number of required anchor measurements, and subsequently, potentially improve energy efficiency, decrease signaling overhead for increased spectral efficiency, reduce latency, and lower the computational load, which are all essential performance criteria in IoT systems [11]. The proposed measurement selection method utilizes the prior information of an underlying tracking method, such as extended Kalman filter (EKF), and the measurement-associated information from potential anchors to arrange the anchors in order of achievable positioning error. This allows harnessing and utilizing all the information available in the measurements, including the measurement geometry, representing thus a more elaborate approach compared to the SNR-based reference scheme. The proposed method is described in both a generic form, enabling utilization of variable tracking configurations and optimization criteria, and in a simplified form, focusing on 2D positioning scenarios with low computational load. As shown through a vast collection of numerical results, the proposed methods enable considerably improved positioning performance, or alternatively utilizing less anchors to reach the same performance compared to the existing SNR-based reference method.

II. DECT-2020 NR AND SYSTEM MODEL

A. Physical-Layer Numerology

The DECT-2020 NR physical layer supports 17 different frequency bands below 6 GHz frequencies [12, table 5.2-1] with a dedicated DECT band located around 1.9 GHz carrier frequency. The specified orthogonal frequency-division

multiplexing (OFDM)-based waveform supports a variable subcarrier spacing (SCS), containing 27 kHz, 54 kHz, 108 kHz, and 216 kHz options [13, Table 4.3-1]. Depending on the used SCS, the achievable nominal bandwidth varies from 1.728 MHz (with 27 kHz SCS) to 221.184 MHz (with 216 kHz SCS). Furthermore, multiple-input multiple-output (MIMO) transmissions are supported with 2, 4, and 8 spatial streams [13, Table 6.3.2-1], which also correspond to a minimum number of required antennas and associated digital transmission chains.

B. Positioning Measurement Model

In this paper, we consider positioning using AoA and ToA-based range measurements obtained from a set of anchors within the signal coverage. For a given user position $\mathbf{p}[n] = [x[n] \ y[n]]^T$, where $x[n]$ and $y[n]$ are the x and y coordinates of the user device at n^{th} time instant, the measurement model is defined as

$$\mathbf{z}[n] = \mathbf{h}(\mathbf{p}[n]) + \mathbf{r}[n], \quad (1)$$

where $\mathbf{z}[n] \in \mathbb{R}^N$ represents measurements from N anchors, $\mathbf{h}: \mathbb{R}^2 \rightarrow \mathbb{R}^N$ is a non-linear measurement model function, and $\mathbf{r}[n] \sim \mathcal{N}(0, \Sigma[n])$ is measurement noise with covariance $\Sigma[n] \in \mathbb{R}^{N \times N}$. For a range measurement, the i^{th} element of the measurement model $\mathbf{h}(\mathbf{p}[n])$ is defined as

$$h_{\text{range},i}(\mathbf{p}[n]) = \|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\|, \quad (2)$$

and for an AoA measurement, obtained at the anchor node, as

$$h_{\text{AoA},i}(\mathbf{p}[n]) = \arctan\left(\frac{y[n] - y_{\text{AN},i}[n]}{x[n] - x_{\text{AN},i}[n]}\right), \quad (3)$$

where $\mathbf{p}_{\text{AN},i}[n] = [x_{\text{AN},i}[n] \ y_{\text{AN},i}[n]]^T$ is the known position of the i^{th} measurement anchor, observed at the n^{th} time instant. In this paper, we study both range and AoA measurements, but do not consider the measurements jointly. However, the presented models enable straightforward implementation for joint measurements, as well as extensions to other types of measurements by modifying the measurement model.

III. PERFORMANCE BOUNDS

A. AoA and Range Estimation Bounds

For revealing the potential AoA and range estimation accuracy with the considered DECT-2020 NR system parameters, we address and express the corresponding Cramér-Rao Lower Bounds (CRLBs). Regarding antenna array based AoA estimation, the CRLB for the estimator variance (in radians) can be expressed as [14]

$$\sigma_{\text{AoA}}^2 \geq \frac{6\lambda^2}{4\gamma\pi^2 L(L^2 - 1)d_{\text{ant}}^2 \cos^2(\theta)}, \quad (4)$$

where L is number of antenna array elements, γ is the SNR, d_{ant} is the antenna spacing, and θ is the true AoA.

For obtaining range measurements, we consider estimation of radio propagation delay through ToA measurements. Assuming the OFDM waveform with uniformly distributed energy

between the subcarriers, the CRLB for ToA-based ranging can be written as [14]

$$\sigma_{\text{range}}^2 \geq \frac{3c^2}{8\gamma\pi^2 f_{\text{SC}}^2 M(M+1)(2M+1)}, \quad (5)$$

where f_{SC} is the subcarrier spacing. Moreover, M is the number of subcarriers at one side of the spectrum, excluding the subcarrier zero so that the overall number of subcarriers is $2M+1$, thus resulting in bandwidth of $B_T = f_{\text{SC}}(2M+1)$.

B. Fisher Information and Measurement-based PEB

Theoretically achievable positioning accuracies can also be analyzed through the corresponding estimator CRLB. A theoretical lower-bound for the covariance matrix of an unbiased estimator $\hat{\mathbf{p}}[n]$ can be formally written as [15]

$$\begin{aligned} \text{CRLB}(\mathbf{p}[n]) &= \mathbf{J}^{-1}(\mathbf{p}[n]) \\ &\leq \mathbb{E}_{\hat{\mathbf{p}}[n]} \{(\hat{\mathbf{p}}[n] - \mathbf{p}[n])(\hat{\mathbf{p}}[n] - \mathbf{p}[n])^T\}, \end{aligned} \quad (6)$$

where $\mathbf{J}(\mathbf{p}[n]) \in \mathbb{R}^{2 \times 2}$ is the Fisher Information Matrix (FIM), and $\mathbb{E}_{\hat{\mathbf{p}}[n]} \{\cdot\}$ denotes the expectation with respect to $\hat{\mathbf{p}}[n]$. Based on the FIM, the positioning error bound (PEB) can now be written as [16]

$$\begin{aligned} \text{PEB}(\mathbf{p}[n]) &= \sqrt{\text{trace}(\mathbf{J}^{-1}(\mathbf{p}[n]))} \\ &= \sqrt{\text{CRLB}(x[n]) + \text{CRLB}(y[n])} \end{aligned} \quad (7)$$

where the PEB represents the lower bound of the positioning root mean squared error (RMSE).

Fisher information that is obtained from a set of measurements $\mathbf{z}[n]$, such as in (1), can be defined through a measurement-based FIM $\mathbf{J}_{\text{meas}}(\mathbf{p}[n]) \in \mathbb{R}^{2 \times 2}$, whose element at the i^{th} row and j^{th} column is written as

$$[\mathbf{J}_{\text{meas}}(\mathbf{p}[n])]_{i,j} = -\mathbb{E}_{\mathbf{p}[n]} \left\{ \frac{\partial^2 \ell(\mathbf{z}[n]|\mathbf{p}[n])}{\partial p_i[n] \partial p_j[n]} \right\}, \quad (8)$$

where $\ell(\mathbf{z}[n]|\mathbf{p}[n])$ denotes the log-likelihood function of the measurements given the user position $\mathbf{p}[n]$, and $p_i[n]$ is the i^{th} element of $\mathbf{p}[n]$. Furthermore, by considering the Gaussian measurement model in (1), the measurement-based FIM can be conveniently written as [15]

$$\mathbf{J}_{\text{meas}}(\mathbf{p}[n]) = \mathbf{H}(\mathbf{p}[n])^T \Sigma^{-1}[n] \mathbf{H}(\mathbf{p}[n]) \quad (9)$$

where $\mathbf{H}(\mathbf{p}[n]) \in \mathbb{R}^{N \times 2}$ is the Jacobian matrix of the measurement function $\mathbf{h}(\mathbf{p}[n])$, defined with respect to $\mathbf{p}[n]$. Assuming uncorrelated measurements from separate anchors, the measurement covariance matrix becomes diagonal, and $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_N^2)$. Accordingly, the measurement-based Fisher information can be presented as a sum of Fisher information of individual measurements as

$$\begin{aligned} \mathbf{J}_{\text{meas}}(\mathbf{p}[n]) &= \sum_{i=1}^N \mathbf{J}_{\text{meas},i}(\mathbf{p}[n]) \\ &= \sum_{i=1}^N \frac{1}{\sigma_i^2[n]} \mathbf{H}_i(\mathbf{p}[n])^T \mathbf{H}_i(\mathbf{p}[n]), \end{aligned} \quad (10)$$

where $\mathbf{J}_{\text{meas},i}(\mathbf{p}[n])$ and $\sigma_i^2[n]$ denote the Fisher information and variance of the i^{th} measurement, respectively. Moreover,

$\mathbf{H}_i(\mathbf{p}[n]) \in \mathbb{R}^{1 \times 2}$ is the i^{th} row of the Jacobian matrix $\mathbf{H}(\mathbf{p}[n])$, which can be presented for range and angle measurements, given in (2) and (3), as

$$\mathbf{H}_{\text{range},i}(\mathbf{p}[n]) = \begin{bmatrix} \frac{x[n] - x_{\text{AN},i}[n]}{\|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\|} & \frac{y[n] - y_{\text{AN},i}[n]}{\|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\|} \end{bmatrix} \text{ and} \quad (11)$$

$$\mathbf{H}_{\text{AoA},i}(\mathbf{p}[n]) = \begin{bmatrix} -\frac{y[n] - y_{\text{AN},i}[n]}{\|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\|^2} & \frac{x[n] - x_{\text{AN},i}[n]}{\|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\|^2} \end{bmatrix},$$

in respective order.

C. User Mobility and Posterior PEB

In case of measurement time series, where measurements are obtained at certain time intervals, temporal correlation of user position can be exploited to enhance positioning performance through appropriate tracking algorithms. To this end, we define a state-vector of the user device at a given time instant n as

$$\mathbf{s}[n] = [x[n], y[n], v_x[n], v_y[n]]^T \in \mathbb{R}^4, \quad (12)$$

where $x[n]$ and $y[n]$ are the x-coordinate and the y-coordinate of the user device, and $v_x[n]$ and $v_y[n]$ are the velocities of the device in the x and y directions, respectively.

For modeling the mobility, we consider a linear state-transition model between consecutive time-instants so that

$$\mathbf{s}[n] = \mathbf{F}\mathbf{s}[n-1] + \mathbf{q}[n], \text{ where} \quad (13)$$

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_{2 \times 2} & \Delta t \cdot \mathbf{I}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & \mathbf{I}_{2 \times 2} \end{bmatrix} \in \mathbb{R}^{4 \times 4} \quad (14)$$

is the state-transition matrix, $\mathbf{q}[n] \sim \mathcal{N}(0, \mathbf{Q}[n])$ denotes process noise, and Δt defines the time interval between the consecutive time instants. Furthermore, by assuming a constant white-noise acceleration (CWNA) based mobility model [17], the process noise covariance matrix is defined as

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^3}{3} \mathbf{I}_{2 \times 2} & \frac{\Delta t^2}{2} \mathbf{I}_{2 \times 2} \\ \frac{\Delta t^2}{2} \mathbf{I}_{2 \times 2} & \Delta t \mathbf{I}_{2 \times 2} \end{bmatrix}, \quad (15)$$

where σ_a^2 is the power spectral density of device acceleration.

With a tracking-based estimator, such as the EKF presented in IV-A, the mobility model in (13) is exploited to provide a priori information on top of measurement information as [18]

$$\mathcal{J}_{\text{post}}(\mathbf{s}[n]) = \mathcal{J}_{\text{prior}}(\mathbf{s}[n]) + \mathcal{J}_{\text{meas}}(\mathbf{s}[n]), \quad (16)$$

where $\mathcal{J}_{\text{post}}(\mathbf{s}[n])$ is the FIM of the a posteriori estimate, $\mathcal{J}_{\text{prior}}(\mathbf{s}[n])$ is the FIM of the a priori estimate, and $\mathcal{J}_{\text{meas}}(\mathbf{s}[n]) = [\mathbf{J}_{\text{meas}}(\mathbf{p}[n]) \quad \mathbf{0}_{2 \times 2}]$ is the FIM obtained from the measurements according to (9). Moreover, the prior information of the state can be formally described as

$$[\mathcal{J}_{\text{prior}}(\mathbf{s}[n])]_{i,j} = -\mathbb{E}_{\mathbf{s}} \left\{ \frac{\partial^2 \log \mathcal{P}(\mathbf{s}[n])}{\partial s_i[n] \partial s_j[n]} \right\}, \quad (17)$$

where $\mathcal{P}(\mathbf{s}[n])$ denotes the prior probability distribution of state $\mathbf{s}[n]$. However, assuming the mobility model in (13), the a priori FIM can be derived from the a posteriori estimate of the previous time step as

$$\mathcal{J}_{\text{prior}}(\mathbf{s}[n]) = (\mathbf{F} \mathcal{J}_{\text{post}}^{-1}(\mathbf{s}[n-1]) \mathbf{F}^T + \mathbf{Q}[n])^{-1}. \quad (18)$$

Algorithm 1 Generic FIM-based measurement selection

Input: Potential measurement-wise FIMs $\mathcal{J}_{\text{meas},i} = \text{blkdiag}(\mathbf{J}_{\text{meas},i}(\mathbf{s}^-[n]), \mathbf{0}_{2 \times 2})$ for $i \in \Omega_{\text{potential}}$, a priori FIM $\mathcal{J}_{\text{prior}} = (\hat{\mathbf{P}}^-[n])^{-1}$, and the number of measurements to be chosen N_{chosen} .^a

Output: Set of chosen measurement indices Ω_{chosen} .

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1: Initialize with empty set:  $\Omega_{\text{chosen}} = \emptyset$ 
2: while  $|\Omega_{\text{chosen}}| \leq N_{\text{chosen}}$  do
3:   Compute Fisher information for  $i \notin \Omega_{\text{chosen}}$ :
4:    $\mathcal{J}_{\text{post},i} \leftarrow [\mathcal{J}_{\text{prior}} + \sum_{k \in \Omega} \mathcal{J}_{\text{meas},k}] + \mathcal{J}_{\text{meas},i}$ 
5:   Compute squared a posteriori PEBb:
6:    $\sigma_{\text{PEB},i}^2 \leftarrow \text{trace}([\mathcal{J}_{\text{post},i}]_{1:2,1:2}^{-1})$ 
7:   Choose measurement with the minimum PEB:
8:    $\hat{i} \leftarrow \arg \min_i \{\sigma_{\text{PEB},i}^2\}$ 
9:   Add  $\hat{i}$  to the chosen measurement set:
10:   $\Omega_{\text{chosen}} \leftarrow \Omega_{\text{chosen}} \cup \{\hat{i}\}$ 
11: end while
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^aBesides the state vector definition in (12), this formulation enables utilization of an arbitrarily chosen state vector.

^bIn addition to, or instead of, the PEB, other tracked state parameters can be considered in this metric, such as velocity, acceleration, and clock bias.

For practical evaluations, at the first time index the a priori FIM can be defined as a diagonal matrix with close to zero diagonal elements, corresponding to very large uncertainty of the state estimate. The bounds will be numerically evaluated in Section V while also serve as inputs to the proposed measurement selection methods, described next.

IV. PROPOSED METHODS

A. EKF-based Tracking and Extraction of Prior Information

Based on the measurement model in (1), and the mobility model in (13), an EKF-based tracking, iterating between a prediction step and an update step, can be utilized for practical position estimation [19]. In the prediction step at time instant n , the mean and covariance of the a priori estimate of the state $\mathbf{s}[n]$ are defined as

$$\begin{aligned} \hat{\mathbf{s}}^-[n] &= \mathbf{F} \hat{\mathbf{s}}^+[n-1] \in \mathbb{R}^4 \\ \hat{\mathbf{P}}^-[n] &= \mathbf{F} \hat{\mathbf{P}}^+[n-1] \mathbf{F}^T + \mathbf{Q}[n] \in \mathbb{R}^{4 \times 4}, \end{aligned} \quad (19)$$

respectively, where $\hat{\mathbf{s}}^+[n-1] \in \mathbb{R}^4$ and $\hat{\mathbf{P}}^+[n-1] \in \mathbb{R}^{4 \times 4}$ are the mean and covariance of the a posteriori estimate of the previous time instant. In this paper, the *prediction step* is in an essential role in the proposed measurement selection approach, utilizing available prior information at each time step. By observing the analogy between the definition of prior information in (18) and the predicted state covariance in (19), estimated prior information at time instant n can be obtained through the inverse of the predicted state covariance as [18]

$$\hat{\mathcal{J}}_{\text{prior}}(\hat{\mathbf{s}}^-[n]) = (\hat{\mathbf{P}}^-[n])^{-1}. \quad (20)$$

After the prediction step, by utilizing a set of obtained measurements $\mathbf{z}[n]$ the a posteriori estimates of the mean and covariance can be acquired through the update step as

$$\begin{aligned} \hat{\mathbf{s}}^+[n] &= \hat{\mathbf{s}}^-[n] + \mathbf{K}[n](\mathbf{z}[n] - \mathbf{h}([\hat{\mathbf{s}}^-[n]]_{1:2})) \\ \hat{\mathbf{P}}^+[n] &= (\mathbf{I}_4 - \mathbf{K}[n] \mathbf{H}[n]) \hat{\mathbf{P}}^-[n], \end{aligned} \quad (21)$$

Algorithm 2 Simplified measurement selection with 2D PEB

Input: Potential measurement-wise FIMs $\mathbf{J}_{\text{meas},i} = \mathbf{J}_{\text{meas},i}(\hat{\mathbf{s}}^-[n])$ for $i \in \Omega_{\text{potential}}$, a priori FIM $\mathbf{J}_{\text{prior}} = [(\hat{\mathbf{P}}^-[n])^{-1}]_{1:2,1:2}$, and the number of chosen measurements N_{chosen} .

Output: Set of chosen measurement indices Ω_{chosen}

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1: Initialize with empty set:  $\Omega_{\text{chosen}} = \emptyset$ 
2: while  $|\Omega_{\text{chosen}}| \leq N_{\text{chosen}}$  do
3:   Compute for  $i \notin \Omega_{\text{chosen}}$ :
4:    $S_{11,i} \leftarrow [\mathbf{J}_{\text{prior}}]_{1,1} + [\mathbf{J}_{\text{meas},i}]_{1,1} + \sum_{k \in \Omega} [\mathbf{J}_{\text{meas},k}]_{1,1}$ 
5:    $S_{12,i} \leftarrow [\mathbf{J}_{\text{prior}}]_{1,2} + [\mathbf{J}_{\text{meas},i}]_{1,2} + \sum_{k \in \Omega} [\mathbf{J}_{\text{meas},k}]_{1,2}$ 
6:    $S_{22,i} \leftarrow [\mathbf{J}_{\text{prior}}]_{2,2} + [\mathbf{J}_{\text{meas},i}]_{2,2} + \sum_{k \in \Omega} [\mathbf{J}_{\text{meas},k}]_{2,2}$ 
7:   Compute squared a posteriori PEB:
8:    $\sigma_{\text{PEB},i}^2 \leftarrow \frac{S_{11,i} + S_{22,i}}{S_{11,i}S_{22,i} - S_{12,i}^2}$ 
9:   Choose measurement with the minimum bound:
10:   $\hat{i} \leftarrow \arg \min_i \{\sigma_{\text{PEB},i}^2\}$ 
11:  Add  $\hat{i}$  to the chosen measurement set:
12:   $\Omega_{\text{chosen}} \leftarrow \Omega_{\text{chosen}} \cup \{\hat{i}\}$ 
13: end while

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respectively, where $\mathbf{K}[n] = \hat{\mathbf{P}}^-[n]\mathbf{H}[n]^T(\mathbf{H}[n]\hat{\mathbf{P}}^-[n]\mathbf{H}[n]^T + \Sigma[n])^{-1} \in \mathbb{R}^{4 \times N}$ is the Kalman gain.

B. Proposed FIM-based Measurement Selection

As shown in (10), each potential anchor measurement is capable of accumulating different amount of Fisher information to the positioning task. Thus, when the number of utilized anchors is restricted, for example, to save energy, it is essential to select the measurements which offer the highest information.

To this end, the proposed measurement selection method utilizes the prior information provided by an appropriate tracking method, such as the EKF presented in Section IV-A. As visible in (10) and (11), evaluating the amount of information offered by each measurement requires knowledge on measurement variances, positions of potential anchors, and the user position at the intended time of measurements. From these, measurement variances are part of standard EKF parameterization and can be estimated, for example, through CRLB evaluations. In addition, anchor positions are assumed to be known as part of the considered positioning framework, and the estimate of user position is directly available by the EKF prediction state, as given in (19).

For any time instant n , the first step in the proposed measurement selection method is to utilize the above-discussed knowledge to evaluate the measurement-associated FIMs for potential anchor measurements, as described in (10). The set of potential anchors, denoted as $\Omega_{\text{potential}}$, include all anchors within the signal coverage that are able to provide positioning measurements. However, minimizing the PEB solely based on the measurement-associated FIMs does not minimize the a posteriori PEB which is also affected by the a priori information, as shown in (16). To this end, the second step in the proposed measurement selection method is to estimate the a priori information by utilizing the predicted state covariance according to (20). Finally, considering a desired number of chosen measurements N_{chosen} , the final set of chosen anchors, denoted as Ω_{chosen} with $|\Omega_{\text{chosen}}| = N_{\text{chosen}}$, can be determined by iteratively accumulating measurement-

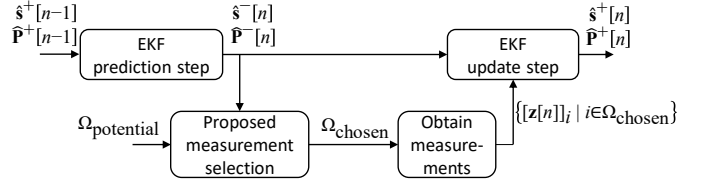


Fig. 1. Illustration of the tracking process including the prediction step, the update step, and the proposed FIM-based measurement selection.

wise a posteriori information according to minimum PEB. Alternatively, it would be theoretically possible to evaluate the a posteriori FIM for all possible measurement combinations. However, in such approach the number of evaluated combinations grows rapidly with the number of anchors, resulting in poor scalability and unfeasible computational burden.

A generic form of the proposed measurement selection method is presented in Algorithm 1. Besides minimizing PEB, the presented algorithm can be modified to minimize errors of any state vector parameters with possible parameter-wise error weighting. Nonetheless, when focusing on the 2D positioning approach considered in this paper, it is possible to simplify the processing steps by utilizing properties of 2×2 matrix inversion, while working only with position-related submatrices of FIMs. The description of the simplified measurement selection approach with 2D positioning focus is presented in Algorithm 2. For a 3D positioning scenario, it is possible to extend the Algorithm 2, or utilize Algorithm 1 with a 3D-compatible state vector. Finally, an illustration of the tracking process utilizing the proposed measurement selection method is shown in Fig. 1, where the block of proposed measurement selection involves processing described in Algorithm 1 or Algorithm 2.

V. NUMERICAL RESULTS AND ANALYSIS

A. Simulation Parameters and Assumptions

For all simulations we consider transmission at the dedicated DECT-2020 NR carrier frequency of $f_c = 1.9$ GHz and SCS of $f_{\text{SC}} = 27$ kHz [1], [20]. Except for ranging accuracy results with variable bandwidth, we fix the number of active subcarriers to 225, corresponding to transmission bandwidth of $B_T = 6.075$ MHz. For AoA estimation, each anchor is equipped with $L = 4$ antennas as the baseline, by using half wavelength antenna spacing $d_{\text{ant}} = c/(2f_c)$, while the cases of $L = 2, 8$ are also considered. The above physical layer configurations follow the DECT-2020 NR specifications in [13].

We assume a circular operation area with 690 m radius, where 1000 anchors are randomly deployed based on uniform distribution. The resulted anchor average density of 669 km^{-2} follows roughly the routing device densities reported in [20], [21]. The AoA and ToA-based range measurements are simulated as zero-mean Gaussian random variables based on the variances given in (4) and (5), respectively. In addition, we include a random Gaussian clock error with 3 ns standard deviation to consider synchronization limitations regarding the ToA measurements. In order to define the SNRs for

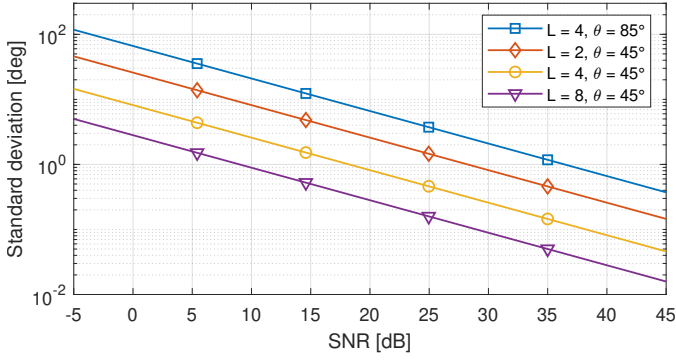


Fig. 2. CRLBs for AoA estimation as a function of SNR with the considered DECT-2020 NR antenna configurations.

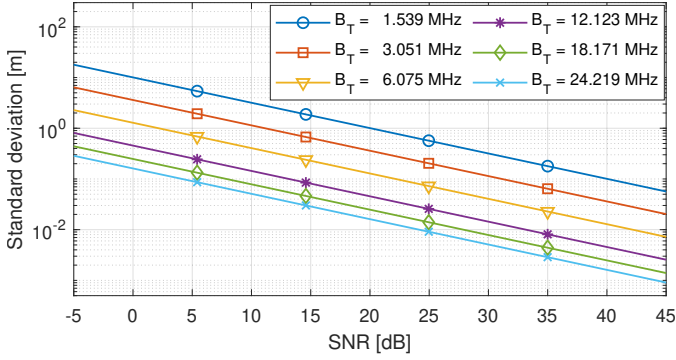


Fig. 3. CRLBs for ToA-based range estimation as a function of SNR with the considered DECT-2020 NR nominal bandwidths.

measurement error variances, we utilize the urban micro path loss model [22, Table 7.4.1-1], while considering only measurements from line-of-sight (LoS) links according to probabilities given in [22, Table 7.4.2-1]. In practice, multipath propagation and inaccuracies in LoS detection can induce outliers in AoA and ToA estimation, in which case such outliers can be managed, for example, using the Bayesian information criterion [23]. Taking into account a possible need for positioning related signaling between the user device and anchors, SNR of $\gamma > 5$ dB is required for the potential anchors to provide measurements.

The results are evaluated over 100 separate user tracks, each track including 10000 time steps, with 10 independent realizations of measurements and anchor deployments. The user device moves with 1 m/s velocity in the operation area with the average power spectral acceleration of $\sigma_a^2 = 12.2 \cdot 10^{-6} \text{ m}^2/\text{s}^3$. The results are evaluated for different measurement time intervals Δt , including 0.1 s, 1 s and 10 s. The initial user position is estimated using a non-linear least squares method.

B. Numerical Results

In Fig. 2, based on (4), potential AoA estimation accuracies for the fundamental DECT-2020 NR MIMO configurations, including 2, 4 and 8 antennas, are shown as a function of SNR, while also illustrating the effect of true AoA on the accuracy with $L = 4$. Regarding range estimation, corresponding poten-

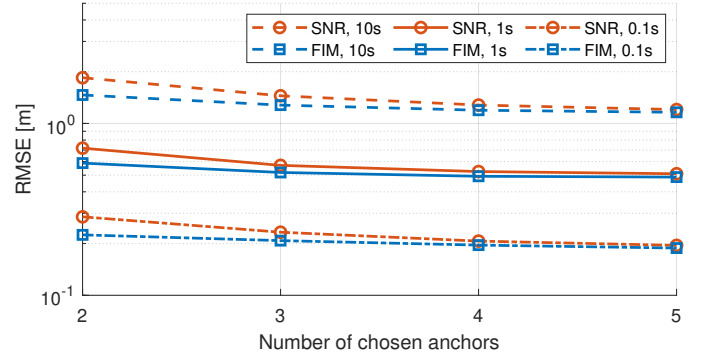


Fig. 4. RMSE as a function of number of chosen anchors for the proposed EKF using FIM and SNR based measurement selection with AoA measurements.

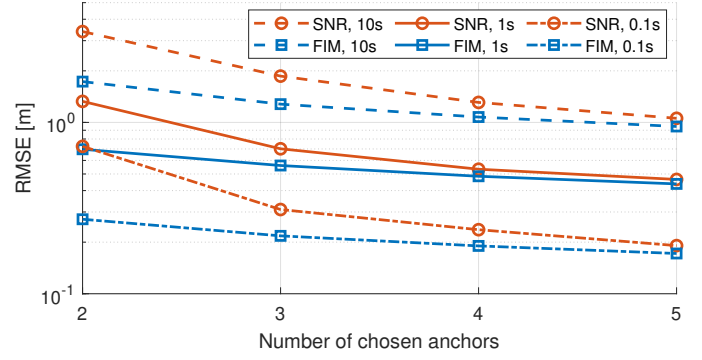


Fig. 5. RMSE as a function of number of chosen anchors for the proposed EKF using FIM and SNR based measurement selection with range measurements.

TABLE I
PERCENTAGE CHANGE OF POSITIONING ERROR RMSES BETWEEN THE TWO ANCHOR SELECTION METHODS, FIM-BASED VS SNR-BASED METHOD.

	AoA			Range		
	0.1 s	1 s	10 s	0.1 s	1 s	10 s
2	-21.5%	-18.2%	-20.7%	-62.5%	-47.5%	-49.0%
3	-10.7%	-9.1%	-11.7%	-29.8%	-20.1%	-31.4%
4	-5.2%	-6.1%	-6.7%	-19.6%	-9.1%	-17.8%
5	-3.5%	-4.4%	-3.6%	-10.0%	-5.8%	-10.4%

tial ranging accuracies with the considered DECT-2020 NR numerology are illustrated in Fig. 3 over different bandwidths B_T according to (5).

The performance of the proposed measurement selection method is compared with a conventional SNR-based selection method, where the anchors are selected according to the highest SNR. Although this approach enables utilization of the highest accuracy measurements, it fails to consider the overall measurement geometry, thus neglecting the available information available in the measurements.

In Fig. 4, the RMSE as a function of the number of chosen anchors with AoA-based positioning is shown for the proposed FIM-based anchor selection and the SNR-based reference method, while considering different measurement time intervals. As expected, the accuracy can be considerably improved by reducing the measurement time interval, which especially helps in tracking the user in high-curvature tracks.

Furthermore, it can be seen that proposed FIM-based anchor selection method enables better positioning accuracy than the SNR-based selection. For example, utilizing 2 anchors with 1 s tracking interval, positioning error RMSE is 18.2% smaller with FIM-based (0.59 m) compared to SNR-based (0.72 m) anchor selection. The difference in positioning accuracy between the FIM and SNR based methods decreases when increasing the number of used anchors.

Corresponding results for positioning with ToA-based ranging are presented in Fig. 5. It can be seen that the trend in results is similar to the AoA case, but the differences between FIM and SNR based methods are more significant. For example, with 2 selected anchors and 1 s tracking interval, RMSE is 47.5% smaller with FIM method (0.70 m) than with SNR method (1.33 m). Again, the difference between the FIM and SNR methods reduces with higher number of selected anchors. This is intuitive, as by increasing the number of anchors there is a diminishing probability for missing the highest information measurements, whereas with small number of anchors the selection becomes more critical.

In order to better compare the performance between the two anchor selection methods, Table I presents the percentage change of positioning RMSEs for using the proposed FIM-based selection method with respect to the SNR-based method. In the case of AoA measurements and 2 selected anchors, positioning RMSE is approximately 20% smaller with FIM-based selection compared to SNR-based selection, and approximately 10% smaller with 3 selected anchors. Moreover, the advantage of FIM-based selection over the SNR-based is further enhanced when using ToA-based range measurements, where with 2 selected anchors the positioning RMSE is from 47.5% to 62.5% smaller with FIM-based selection than SNR-based selection. Moreover, when using 3 anchors, the RMSE is reduced from 20.1% to 31.4% with the FIM-based approach, and even with 5 used anchors the difference is still from 5.8% to 10.4% in favor of the FIM-based approach.

In some use cases it is important to avoid large errors and provide consistent positioning performance over time. For this purpose, in Table II, the 99.9th percentile of positioning errors by utilizing 2 anchors are shown for the FIM and SNR based anchor selection methods, while considering both AoA and range measurements as well as different tracking intervals. The results show that in terms of 99.9th percentile of positioning errors the FIM-based anchor selection method clearly outperforms the SNR-based method with both AoA and ToA based measurements. For example, with range measurements and 1 s tracking interval, the 99.9th percentile of positioning error for the FIM-based and SNR-based selection method are 0.84 m and 6.13 m, respectively, indicating approximately 86% smaller error with the FIM-based method.

Finally, in Fig. 6, the RMSE as a function of time instant, considering 1 s measurement interval and 2 selected anchors, is illustrated for the EKF-based tracking, as well as for the a posteriori PEB, evaluated according to Fisher information defined in (16). In the figure, the filtering gain, achieved by exploiting the prior information stemming from the mobility

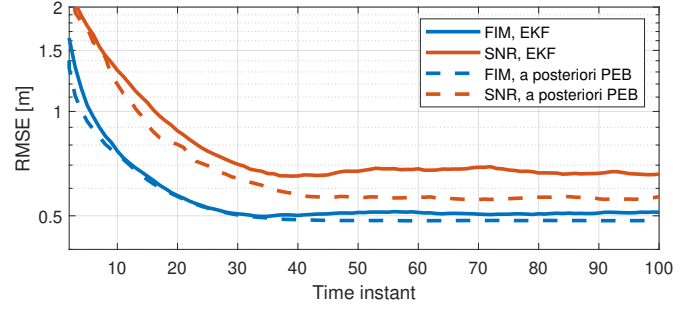


Fig. 6. RMSE as a function of time instants for FIM and SNR based measurement selection, considering EKF-based estimation and corresponding a posteriori bounds.

TABLE II
99.9TH PERCENTILES OF POSITIONING ERRORS IN METERS USING EKF TRACKING WITH 2 SELECTED ANCHORS

	AoA		Range	
	FIM	SNR	FIM	SNR
0.1 s	1.1266	1.6060	0.8422	6.1330
1 s	3.2499	4.4762	2.5995	16.9349
10 s	10.3525	14.8658	12.0734	35.4046

model, is seen as decreasing RMSE trend in the beginning of the process. It can be seen that the used EKF-based tracking approach achieves close to PEB performance, especially considering the proposed FIM-based selection method which clearly outperforms the SNR-based benchmark method.

VI. CONCLUSION

In this work, considering the IoT framework with a specific focus on the DECT-2020 NR standard, we proposed a method for selecting the most important positioning measurements to avoid utilizing and processing negligible measurements, thus reducing energy consumption, latency and computational load. Moreover, exploiting measurement-based Fisher information from potential anchor measurements, and the information provided by the prediction step of a tracking algorithm according to the underlying mobility model, the proposed measurement selection method were used to minimize the a posteriori PEB. The proposed measurement selection method was described in a generic form, enabling use of arbitrary tracking state variables with variable performance criteria, as well as in a simplified form, considering 2D positioning with reduced computational load.

With the provided numerical results, first, positioning measurement accuracies for range and angle measurements were evaluated under DECT-2020 NR system parameters. After this, the performance of the proposed measurement selection approach was assessed and compared against the benchmark method with SNR-based measurement selection, while also addressing the corresponding theoretical bounds. The results show that the proposed method outperforms the SNR-based solution, especially with range measurements, and a small number of used measurements. Furthermore, the proposed method significantly reduces the largest positioning errors, which can be a crucial asset for several IoT use cases.

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